



Capital University of Science and Technology
Department of Software Engineering

SE2512: Digital Logic Design
Midterm Exam

Semester: Spring 2025

Date: May 05, 2025

Max Marks: 20

Time: 90 Min

Instructor: Dr. Muhammad Riaz (S-1, S-2), Fouzia Qureshi (S3, S4)

Read the following instructions carefully:

1. Calculators are not allowed.
3. In your solutions, show all details you claim credit for.
4. There are FOUR questions in this paper on TWO pages. Attempt all questions.
5. This exam carries 20% weightage towards the final evaluation

Name: _____ Reg. No. _____ Section: _____

Part 1 (CLO-1)

(10 marks)

Question 01

- (a) Convert the binary number 10001.0101 to decimal. Show your steps. Show only the first two digits after the decimal point.
- (b) Convert the decimal numbers 47.4375 to binary: (Show your steps.)

(2+3)

Question 02

- (a) Given the two decimal numbers $X = 72532$ and $Y = 3250$, perform the subtraction $X - Y$ and $Y - X$ using 10's complement.
- (b) Represent the unsigned decimal numbers 609 and 516 in BCD, and then show the steps necessary to form their sum.

(2+3)

Part 2 (CLO-2)

(10 marks)

Question 03

- (a) Simplify the following Boolean expression to a minimum number literals using **algebraic manipulation**:

$$F = a'bc + abc' + abc + a'bc'$$

- (b) Draw the logic diagram of the following expression using **NAND** and **NOT** gates only.

$$F = x(y'z' + yz) + xyz'$$

(2+3)

Question 04

- (a) For a given Boolean function $F(w, x, y, z) = \Sigma(1, 4, 5, 6, 12, 14, 15)$ which has the do not care conditions of $d(w, x, y, z) = \Sigma(3, 7, 11)$, determine simplified sum of products using K-map.

- (b) Draw the truth table of $F = x(y'z' + yz) + x'yz'$

(2+3)

Good Luck